

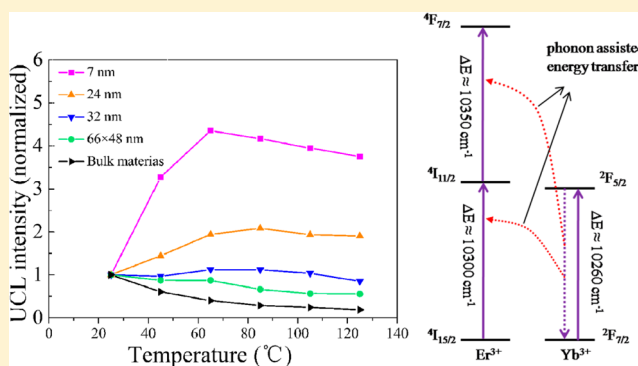
Anomalous Temperature-Dependent Upconversion Luminescence of Small-Sized $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ Nanoparticles

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S Supporting Information

ABSTRACT: Size-dependent quantum confinement has important effects on the energy transfer and radiative and nonradiative transitions in nanophosphors. For lanthanide-doped nanoparticles, the confinement effect is induced mostly via electron–phonon interaction, and analysis of temperature-dependent spectroscopic properties provides an effective method for disclosing its underlying mechanism. Herein, an intriguing and unprecedented enhancement of the upconversion luminescence (UCL) at higher temperatures in hexagonal-phase $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ upconversion nanoparticles (UCNPs) is reported. Moreover, this anomalous UCL enhancement shows a strong dependence on the particle size and becomes more significant for UCNPs with a smaller size. This anomalous thermal behavior is interpreted on the basis of phonon-assisted energy transfer and phonon confinement effect. The findings are relevant to the engineering of the nanostructures of UCNPs and to the further understanding of the UCL mechanism.



INTRODUCTION

Upconversion nanoparticles, that can emit visible light under near-infrared (NIR) excitation, have attracted great attention for their potential applications in a variety of fields, such as biomedical imaging, labeling, and photodynamic therapy.^{1–5} When used as biomedical imaging probes, upconversion materials exhibit many unique advantages over conventional down-conversion counterparts, including high signal-to-noise ratios, low toxicity and radiation damage, good photostability, and improved tissue penetration depth.^{3–5}

Among various upconversion materials, hexagonal (β)-phase $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ upconversion nanoparticles (UCNPs) have been reported as the most efficient NIR-to-visible materials. However, their absolute quantum yields remain still low and are usually less than 1%,⁶ significantly limiting the use of UCNPs in the biomedical field. Several factors have been found to affect the upconversion luminescence (UCL) efficiency, for example, crystalline phases, dopant concentrations, sensitizer–activator ratios, and surface properties.^{7–10} The development of core–shell structures has proved able to enhance their UCL emissions. However, even for core–shell structured $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ UCNPs (β -phase, ~ 30 nm) with optimized Yb^{3+} and Er^{3+} concentrations, the quantum yield is only about 0.3%.⁶ Further increasing the quantum yields of UCNPs, which is preferred for in vivo applications, still requires a full understanding about energy loss mechanisms in UCL processes of NaYF_4 -based UCNPs. In addition, quantum yields of $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ UCNPs show a significant size-dependence

and become even lower as their sizes reduce.^{6,9–16} However, the mechanism of size-dependent UCL properties is still in debate. For example, the decreasing UCL efficiency with the reduction of particle sizes was explained to be the surface effect^{13,14} and the phonon confinement effect,^{9,15,16} respectively. Liu et al. have suggested that the discreteness of phonon density of states or the absence of low-energy phonons due to the confinement effect may impact on the phonon-assisted energy transfer process and induce the decrease of the UCL intensity of small-sized UCNPs.¹⁵ Lim et al.⁹ and Schietinger et al.¹⁶ have reported that the relative increase in green emission with reducing the particle size can be attributed to the phonon confinement effect in smaller nanoparticles, therefore affecting the nonradiative phonon relaxation. Oppositely, Wang et al.¹³ and Shan et al.¹⁴ have pointed out that the surface quenching effect should be mainly responsible for the size-dependent UCL on the basis of the comparative spectroscopic investigation of UCNPs with or without a surface protection shell and the dynamic probing of the UCL, respectively. There still has no consistent view to adequately explain all the size-dependent UCL phenomena. Therefore, in-depth studies are still needed to validate whether the surface effect or the phonon confinement dominates the size-dependence UCL properties, or these two effects coexist in smaller UCNPs.

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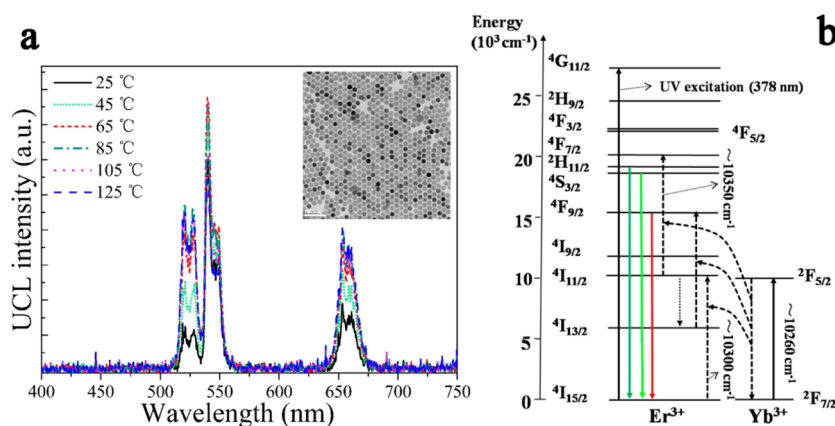


Figure 1. (a) Temperature-dependent UCL spectra of 24 nm NaYF₄:Yb³⁺, Er³⁺ UCNP (inset: the corresponding TEM image). (b) Relevant excitation and luminescence scheme in Yb³⁺/Er³⁺ system: energy transfer (dashed), radiative processes (full), multiphonon relaxation (dotted).

Luminescence is a competitive process between radiative and nonradiative transitions. In order to improve the quantum yields of UCNP, a deep understanding about the nonradiative process is necessary. Analysis of temperature-dependent UCL can provide a useful means for understanding the nonradiative process in UCNP. Temperature-dependent UCL spectra studies may also be helpful for clarifying the roles of the surface effect and the phonon confinement in affecting these size-dependent UCL, on the basis of different temperature-dependent behaviors of the two effects. The intensity ratio of $^2\text{H}_{11/2} \rightarrow ^4\text{I}_{15/2}$ and $^4\text{S}_{3/2} \rightarrow ^4\text{I}_{15/2}$ transitions of NaYF₄:Yb³⁺, Er³⁺ UCNP is sensitive to temperature, and it provides a simple and reliable method for temperature sensing.^{17–19} Studies on their temperature-dependent UCL properties may provide useful information for the practical applications of NaYF₄:Yb³⁺, Er³⁺ UCNP as novel temperature sensing probes.

As mentioned above, the study of temperature-dependent UCL spectra will help with the further understanding of the UCL and size-dependent mechanism, and the use of UCNP as temperature sensing probes. Suyver et al.²⁰ have studied the temperature-dependent UCL of NaYF₄:Yb³⁺, Er³⁺ in the temperature range of 5–200 K, and found an UCL enhancement phenomenon with increasing temperature from 5 to 50 K. Recently, Song et al.²¹ have reported that this UCL enhancement phenomenon exhibited in a cryogenic temperature range depended on the crystalline phase. For cubic (α)-phase NaYF₄:Yb³⁺, Er³⁺ UCNP, the overall UCL intensity decreased solely with increasing temperature, while for β -phase UCNP ranging from several tens to thousands of nanometers the UCL intensity demonstrated a maximum at about 100 K. The temperature-dependent UCL enhancement that exhibited below room temperature has been attributed to the increased population of higher crystal-field component of the $^2\text{F}_{5/2}$ energy level of Yb³⁺ ions, leading to more efficient energy transfer between Yb³⁺ and Er³⁺ ions.^{20,21} Shan et al.¹⁹ and Zhao et al.²² have studied the temperature-dependent UCL spectra of NaYF₄:Yb³⁺, Er³⁺ nanoparticles in the temperature region higher than room temperature (300–450 K), and a general thermal quenching behavior has been observed. So far, due to the difficulty in controlling particle size of β -NaYF₄ UCNP, there still lack of comprehensive data to evaluate the temperature-dependent UCL of β -NaYF₄ UCNP in a broad size range, especially for the particle size approaching 20 nm or smaller.

In this paper, we describe an experimental investigation of temperature-dependent UCL of the β -NaYF₄:Yb³⁺, Er³⁺ UCNP with different particle sizes. An unexpected and anomalous UCL enhancement with increasing temperature was first observed for small-sized UCNP (<30 nm). Systemic analysis reveals that this abnormal behavior arises from the enhanced phonon-assisted energy transfer process at higher temperatures. The result is not only favorable for the engineering of the nanostructures of UCNP, but also provides useful information on the thorough understanding of UCL mechanisms.

EXPERIMENTAL METHODS

Synthetic Procedure. β -NaYF₄:Yb³⁺, Er³⁺ UCNP with different sizes were synthesized according to a modification of the previously reported procedure using rare-earth acetates as precursors.²³ In a typical synthesis of 24 nm nanoparticles, 1 mmol of rare-earth (Re) acetates (Y/Yb/Er = 78:20:2) with 6 mL of oleic acid (OA) and 15 mL of 1-octadecene (ODE) were added to a 100 mL flask to form a mixed solution under vigorous stirring. The solution was heated to 100 °C for 30 min before cooling to 50 °C. Then a methanol solution (20 mL) containing NH₄F (4 mmol) and NaOH (2.5 mmol) was added, and the resulting solution was kept at 50 °C for 30 min. After methanol was evaporated, the solution was heated to 300 °C under an argon atmosphere for 90 min and then cooled down to room temperature. The nanoparticles were precipitated by the addition of ethanol and isolated via centrifugation. The as-precipitated nanoparticles were washed several times with ethanol and dried in air at 70 °C overnight. NaYF₄:Yb³⁺, Er³⁺ UCNP with particle sizes of 32 nm and 66 × 48 nm were prepared in an identical procedure, except for the use of 3.5 and 3 mmol of NH₄F in the synthesis, respectively.

For the synthesis of NaYF₄:Yb³⁺, Er³⁺@NaYF₄ core-shell UCNP, the shell precursor was first prepared by adding yttrium acetate hydrate (0.5 mmol) into OA (6 mL) and ODE (15 mL) and subsequently heating at 100 °C for 30 min. When the solution was cooled down to 50 °C, NaYF₄:Yb³⁺, Er³⁺ core nanoparticles in cyclohexane were added along with a methanol solution (10 mL) of NH₄F (2 mmol) and NaOH (1.25 mmol). The resulting solution was stirred at 50 °C for 30 min and then heated to 300 °C for 90 min. After the solution was cooled down to room temperature, the obtained core-shell nanoparticles were collected and washed by the same procedure as that of the core-only nanoparticles.

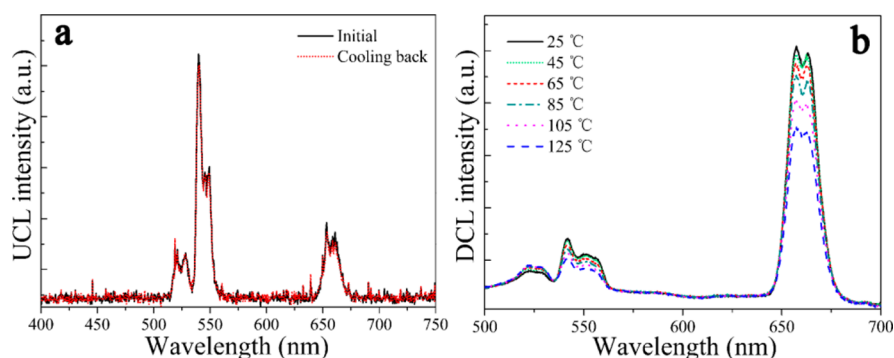


Figure 2. (a) UCL spectra before and after a heating–cooling cycle and (b) temperature-dependent DCL spectra of 24 nm $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ UCNPs.

The synthetic procedure of 7 nm $\text{NaGdF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ UCNPs was the same as that of 24 nm $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ nanoparticles, except that gadolinium acetate instead of yttrium acetate was used as the precursor.

Materials Characterization. The nanoparticle morphology was recorded on a Tecnai G2 transmission electron microscope (TEM) and a Sirion-400 scanning electron microscope (SEM). A temperature-controlled heating system was used for temperature-dependent UCL spectra measurements. The solid powder samples were put into a copper sample cell, which is 10 mm in diameter and 2 mm in depth. A ceramic plate was placed below the sample cell for heating, which was connected to an intelligent PID regulator (LU-906M, Anthone Electronics Co. Ltd.) for precise temperature control. The temperature was monitored by a thermocouple embedded on the copper surface near the sample cell, which promised the readings from the thermocouple were the same as the temperatures of UCNPs. In the measurements of UCL spectra, a continuous 975 nm diode laser was used as the excitation source. The laser power was fixed at about 35 mW, and the beam was directly incident on a spot size of 5 mm in diameter through an optical fiber. The UCL signals were collected and analyzed by a portable spectrometer (Maya2000, Ocean Optics Co.). The temperature-dependent downconversion luminescence (DCL) spectra were obtained by a Hitachi F-7000 fluorescence spectrometer with the help of the same temperature-controlled sample holder. Temperature-dependent diffuse reflection spectra of solid powder samples were measured by using a Cary 5000 UV–vis–NIR spectrophotometer, with the help of an internal diffuse reflectance accessory and a resistance-heating sample cell.

Results and Discussion. Temperature-dependent UCL spectra of typical $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ UCNPs are shown in Figure 1a, and the average size of UCNPs was determined to be about 24 nm by TEM analysis (inset of Figure 1a). Under 975 nm laser excitation, green and red upconversion emissions centered at around 525, 545, and 660 nm were observed, corresponding to the $^2\text{H}_{11/2} \rightarrow ^4\text{I}_{15/2}$, $^4\text{S}_{3/2} \rightarrow ^4\text{I}_{15/2}$, and $^4\text{F}_{9/2} \rightarrow ^4\text{I}_{15/2}$ transitions of Er^{3+} ions (Figure 1b), respectively. In the temperature range from 25 to 85 °C, an unexpected UCL enhancement for all the emission bands (525, 545, and 660 nm) was observed with increasing temperature. It is clearly distinguished from the previously reported enhancement of the $^2\text{H}_{11/2} \rightarrow ^4\text{I}_{15/2}$ transition (525 nm) in comparison with the $^4\text{S}_{3/2} \rightarrow ^4\text{I}_{15/2}$ transition (545 nm). The latter phenomenon has been well explained by the existence of a thermal equilibrium between $^2\text{H}_{11/2}$ and $^4\text{S}_{3/2}$ energy levels of Er^{3+} ions, owing to

their narrow energy gap ($\sim 700 \text{ cm}^{-1}$), which has been used to design novel noncontact nanothermometers based on Er^{3+} -activated UCNPs.^{17–19} Herein, an overall increase in UCL intensity with the rise of temperature was first found. The integrated intensity over a wavelength range of 510–680 nm was doubled as the temperature increased from 25 to 85 °C. This result is contradictory to the previous report about 40 nm $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ UCNPs (cubic phase), where a continuous decrease in UCL intensity was observed with increasing temperature.²² This overall increase in the UCL intensity with temperature is also different from that reported by Suyver et al.²⁰ and Song et al.,²¹ where the UCL intensity increased with temperature in the cryogenic region (5–150 K). It implies that a different mechanism dominates this anomalous thermal behavior. In addition, it should be noted that as the temperature was higher than 85 °C, a slight decrease of UCL intensity with temperature was found, which is consistent with the normal thermal quenching behavior of luminescent materials.

Bednarkiewicz et al.²⁴ have reported a giant and irreversible UCL enhancement in ultrasmall $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ nanoparticles, which was attributed to an infrared-laser-induced annealing effect. Figure 2a represents UCL spectra of 24 nm UCNPs before and after a heating–cooling cycle. The UCL intensity nearly recovered to the original intensity after cooling down to room temperature. Therefore, the anomalous UCL enhancement at elevated temperatures cannot be induced by the laser annealing effect. Even for the dispersed UCNPs in hexane solution, a similar increase in the UCL intensity with increasing temperature could also be observed (not shown).

For $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ UCNPs, the efficient UCL has been ascribed to be an energy transfer upconversion (ETU) mechanism,²⁵ taking advantage of good resonances between the $^2\text{F}_{7/2} \rightarrow ^2\text{F}_{5/2}$ transition of Yb^{3+} ions and diverse f-f transitions of Er^{3+} ions (Figure 1b). For the ETU process, the UCL quantum yields depend on the absorption rate, energy transfer rate, and radiative and nonradiative transition rates of activators.^{10,19} No obvious changes can be found for diffuse reflection spectra of $\text{NaYF}_4\text{:Yb}^{3+}, \text{Er}^{3+}$ UCNPs with increasing temperature (see Figure S1 in the Supporting Information), and therefore, the variation in absorption rate can be ruled out as the cause of the anomalous UCL enhancement with temperature. In order to exclude the influences of the radiative and nonradiative transition processes on this abnormal temperature effect, temperature-dependent DCL spectra were also measured under the excitation of the 378 nm light (Figure 2b). Unlike the UCL, the DCL intensity decreased gradually

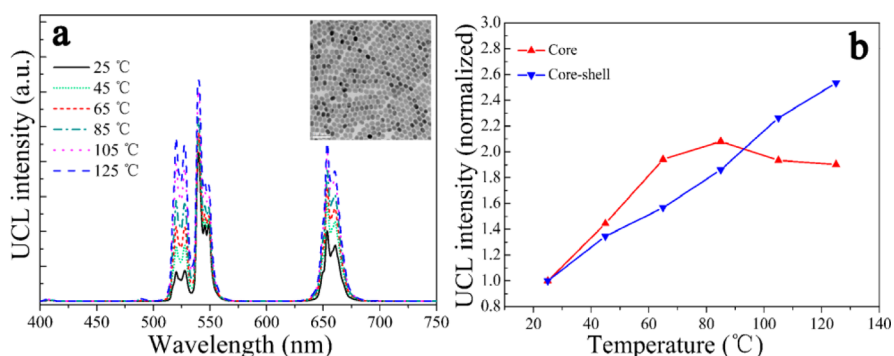


Figure 3. (a) Temperature-dependent UCL spectra of $\text{NaYF}_4\text{:Yb}^{3+}$, $\text{Er}^{3+}\text{@NaYF}_4$ core-shell UCNPs (inset: corresponding TEM image). (b) Integrated UCL intensities of core-only and core-shell UCNPs as a function of temperature. Intensities were calculated by integrating the spectral intensity of the UCL spectra over a wavelength range of 510–680 nm and normalized to that at 25 °C for each sample.

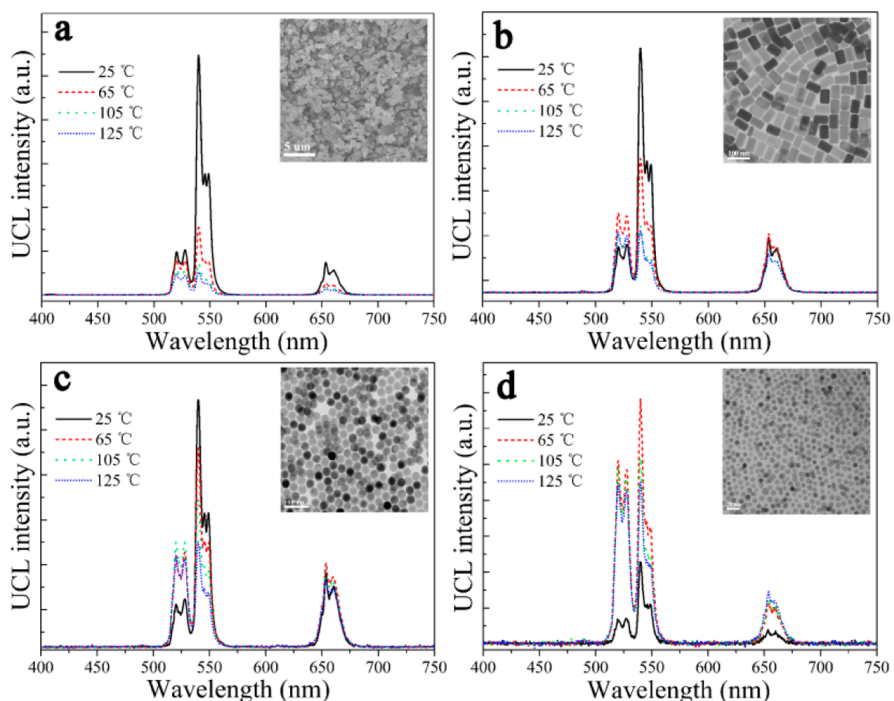


Figure 4. Temperature-dependent UCL spectra of (a) bulk $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} , (b) 66×48 nm $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNPs, (c) 32 nm $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNPs, and (d) 7 nm $\text{NaGdF}_4\text{:Yb}^{3+}$, Er^{3+} UCNPs (insets: corresponding SEM or TEM images).

with increasing temperature. This is consistent with the well-known thermal quenching behavior of luminescent materials, and the nonradiative transition process is promoted at higher temperatures compared with the radiative transition process. The promoted nonradiative decay is responsible for the relative decrease of the UCL intensity in the temperature range higher than 85 °C (Figure 1a); however, it cannot explain the anomalous UCL enhancement in the temperature range of 25–85 °C. The different temperature-dependent behaviors between UCL and DCL are related to their distinctive energy conversion mechanisms. For the UCL process, Yb^{3+} ions act as the sensitizers, which adsorb and transfer the excitation energy to adjacent Er^{3+} ions to populate $^4\text{I}_{11/2}$ and then $^2\text{H}_{11/2}$, $^4\text{S}_{3/2}$, $^4\text{F}_{9/2}$ levels of Er^{3+} ions via the ETU process (Figure 1b). For the DCL process, however, the $^4\text{G}_{11/2}$ energy level of Er^{3+} can be directly excited via absorbing a 378 nm photon without the energy transfer process (Figure 1b). The comparison between UCL and DCL indicates that the energy transfer process is

highly relevant to the abnormal UCL enhancement of $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNPs at higher temperatures.

Due to the large surface-area to volume (SA/Vol) ratio of nanoparticles, most of activators (Er^{3+}) locate at the particle surface, making them susceptible to be affected by surface defects and ligands. In order to probe the surface effects on the temperature-dependent UCL, a comparative investigation of $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} UCNPs with or without a thin surface protection shell was performed. An undoped NaYF_4 shell was grown around the 24 nm $\text{NaYF}_4\text{:Yb}^{3+}$, Er^{3+} core, and the corresponding TEM image is shown in the inset of Figure 3a. The average particles size of core-shell UCNPs is about 28 nm, indicating that the NaYF_4 shell with the thickness of ~ 2 nm was successfully coated. As the temperature increased, a similar increase of the overall UCL intensity was observed for $\text{NaYF}_4\text{:Yb}^{3+}$, $\text{Er}^{3+}\text{@NaYF}_4$ core-shell UCNPs (Figure 3a). It indicates that the surface quenching effect has no direct relationship with the UCL enhancement at higher temperatures. Figure 3b represents the integrated intensity of core-

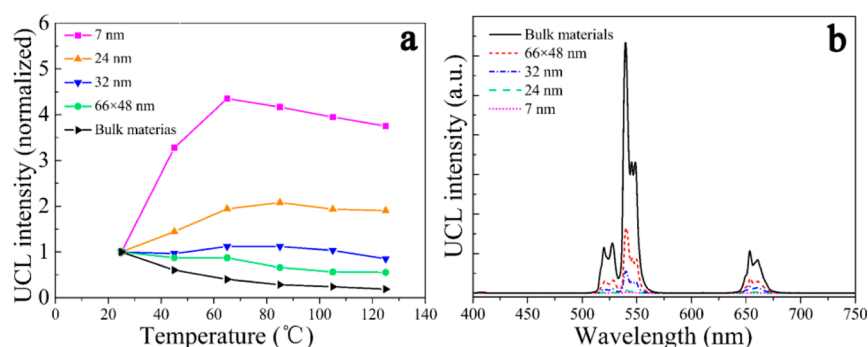


Figure 5. (a) Integrated UCL intensities of different-sized particles as a function of temperature. Integrated intensities at various temperatures were normalized to that at 25 °C for each sample. (b) Room-temperature UCL spectra of Yb³⁺/Er³⁺ codoped particles with various sizes.

only and core-shell UCNPs as a function of temperature. Unlike core-only UCNPs, a continuous increase of the UCL intensity was found for core-shell UCNPs with the increase of the temperature from 25 to 125 °C, which is mainly due to the fact that the shell can effectively suppress the thermal quenching effect arising from the surface-absorbed ligands.

Based on the above results, the laser heating, the nonradiative process, and the surface effect are believed to be insignificant in affecting the anomalous UCL enhancement at higher temperatures. The comparative investigation between temperature-dependent UCL and DCL spectra shows that the energy transfer process is highly relevant to this observation. As shown in Figure 1b, there are slight energy differences (40–90 cm^{−1}) between the ²F_{7/2} → ²F_{5/2} transition (~10260 cm^{−1}) of Yb³⁺ ions and the ⁴I_{15/2} → ⁴I_{11/2} (~10300 cm^{−1}) or ⁴I_{11/2} → ⁴F_{7/2} (~10350 cm^{−1}) transition of Er³⁺ ions.^{26,27} The energy transfer between donors (Yb³⁺) and acceptors (Er³⁺) requires low-energy phonons to match these energy differences.²⁸ When the dimension of the particles is in nanoscale, the phonon density of states becomes discrete and low-energy phonon modes are cutoff. Therefore, phonon-assisted energy transfer processes will be affected because of the lack of low-energy phonons, leading to the decrease of the UCL efficiency in small-sized UCNPs. However, at higher temperatures, the phonon occupation number will increase and the phonon confinement effect can be weakened, leading to the increase of the energy transfer efficiency and the enhancement of the UCL intensity.²⁹ Therefore, the anomalous UCL enhancement can be attributed to the changes of phonon-assisted energy transfer efficiency with temperature due to the phonon confinement effect in small-sized UCNPs.

To further test this hypothesis, temperature-dependent UCL properties of UCNPs with various sizes were studied (Figure 4). As a comparison, temperature-dependent UCL spectra of bulk NaYF₄:Yb³⁺, Er³⁺ materials (purchased from Gona Powder Corporation, China, >300 nm) were also presented (Figure 4a). In order to prepare ultrasmall UCNPs, gadolinium (Gd) was used to replace yttrium (Y) as host elements.³⁰ TEM analysis indicates that NaGdF₄:Yb³⁺, Er³⁺ UCNPs with an average size of 7 nm were successfully prepared (Figure 4d). Figure 5a gives integrated UCL intensities of different-sized UCNPs as a function of temperature. It can be clearly seen that this anomalous UCL enhancement exhibits a strong size dependence. For 32 nm UCNPs, the increase of the UCL intensity with temperature is almost negligible. When the particle size is smaller than 32 nm, the UCL enhancement becomes more significant with the reduction of the particle size, due to the enhanced phonon confinement effect. Especially for

7 nm UCNPs, the integrated intensity increased by ~3.3 times as the temperature increased from 25 to 65 °C. However, when the particle size is larger than 32 nm, only the gradual decreases in the UCL intensity is observed with increasing temperature. For UCNPs with a larger size, the phonon confinement becomes insignificant and there are enough low-energy phonons to compensate the energy differences between Yb³⁺ and Er³⁺ ions.^{15,16} The general thermal quenching mechanism will dominate the changes of UCL spectra with temperature and, therefore, lead to the gradual decrease of the UCL intensity with the increase of temperature. Figure 3b shows that the increase of the UCL intensity for core-shell UCNPs is slightly slower than that of core-only ones in the temperature range of 25–85 °C, which may be also related to the larger size of core-shell UCNPs.

This observation will be also helpful for clarifying the size-dependent UCL mechanism of Yb³⁺/Ln³⁺ (Ln = Er, Ho, Tm) codoped UCNPs. As shown in Figure 5b, under the same excitation conditions, the UCL intensity rapidly decreases with the reduction of the particle size. Wang et al.¹³ have pointed out that the phonon confinement is insignificant and the surface quenching effect dominates in affecting the size-dependent UCL of NaYF₄:Yb³⁺, Tm³⁺, through comparative spectroscopic studies of nanoparticles with and without a protection shell. Based on UCL dynamic spectra, the same conclusion has been made by Shan et al.¹⁴ The above results about temperature-dependent UCL spectra indicate that, besides the surface quenching effect, the phonon confinement effect should not be neglected in explaining the size-dependent UCL of UCNPs. Especially for ultrasmall UCNPs, we believe that the surface effect and the phonon confinement coexist, leading to the rapid decrease of the UCL intensity with reducing particle size. Temperature-dependent UCL dynamic spectra and comparative investigations of UCNPs doped with various Ln³⁺ ions will provide more useful information for further understanding the influences of the phonon confinement on their size-dependent and temperature-dependent spectroscopic properties, which is to be addressed in future studies.

CONCLUSIONS

In summary, temperature-dependent UCL properties of β-NaYF₄:Yb³⁺, Er³⁺ UCNPs with various particle sizes are studied. An anomalous UCL enhancement with the increase of temperature is observed for small-sized UCNPs (<30 nm). Also, this temperature-induced enhancement effect shows a strong size dependence, which is more obvious with decreasing the particle size and becomes insignificant for UCNPs larger

than 30 nm. We excluded that the surface effect, laser annealing, and nonradiative processes should be responsible for this observation and attributed this behavior to the change of phonon-assisted energy transfer processes. For smaller UCNPs, the lack of low-energy phonon modes due to the phonon confinement effect limits the energy transfer between Yb^{3+} and Er^{3+} ions. At higher temperatures, the increase in phonon density can promote the energy transfer efficiency and results in the UCL enhancement with temperature in small-sized UCNPs. The result will further support the understanding of UCL mechanisms in general and can also provide useful information for the pursuit of the better tailored UCL materials.

■ ASSOCIATED CONTENT

● Supporting Information

Temperature-dependent diffuse reflection spectra of upconversion particles with various sizes. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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